Lab2: Spatial domain and frequency domain. Fourier transform and its applications.

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Deadline.

Exercise 1.

Open the image `lena_noisy.png` in the working directory. Write a program that removes the noise and recovers the original clear image. Show the resulting image in the Fourier domain as well.

$\Longrightarrow$

How would the filter change with `lena_verynoisy2.png` image?

Exercise 2.

Take a picture (wherever you want) that contains significant-quick changes in appearance which might be prone to change the image appearance after subsampling (for example, a picture of striped pants, in which thin vertical stripes turn into thick horizontal stripes, or a picture of a brick wall, in which some pattern appears). Try subsampling by different factors: 2, 4, 8,...

$\Longrightarrow$

Answer (in your report, providing examples and visual results) the following questions:

- What kind of pre-processing can be applied to the image so that aliasing artifacts do not appear after subsampling? *Warning: If you think that no pre-processing is required, try to subsample the image of the stripped pants :)*
- What would be the appropriate pre-processing for subsampling by a factor of 2? What about subsampling by a factor of 4?
- Does ringing appear? If so, how could it be avoided?

Exercise 3.

Consider the vectors:

```
matlab⟩ x = [ 0 0 0 0 0 1 1 1 1 1 ]
matlab⟩ y = [ 0 0 0 0 1 0 0 0 0 0 ]
```

and translate them to the right (using the formula explained in class).

What happens if the translation distance is not an integer number (translation to sub-pixel positions)? That is, try to consider the following vectors

```
matlab⟩ x = [ 0 0 0 0 0 1 1 1 1 1 ]
matlab⟩ y = [ 0 0 0 0 1 0 0 0 0 0 ]
```

and do the translation to the right with the following distances: 0.5 pixel, 0.1 pixel.

Repeat the displacement by 0.5 pixel twice, and by 0.1 pixel 10 times. Show all intermediate results and interpret the results.

Remark: recall the exercise 7 of the Seminar 3, or see the page 80 of the notes.

Exercise 4.

Implement a precise *shear* operator, which performs the following affine transformation of the domain of an image:

$$(x, y) \mapsto (x + ay, y)$$

where a is any real number.

Once done, you can implement the method of Yaroslavski to rotate an image (pages 80-81 of the notes or exercise 8 of the Seminar 3). According to this method, a rotation is expressed as the product of three shears. Having obtained the appropriate translations for a an angle equal to $pi/2$ radians, compare the results of this method with the naïve methods for rotations (rotate and interpolate pixel values) or using a drawing program (paint or similar).

$$\begin{pmatrix} cos\,\theta & sin\,\theta \\ -\sin\theta & cos\,\theta \end{pmatrix} = \begin{pmatrix} 1 & -tan\,\theta/2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \sin\theta & 1 \end{pmatrix} \begin{pmatrix} 1 & -tan\,\theta/2 \\ 0 & 1 \end{pmatrix}$$

Optional Exercise.

Taking both *Mcqueen* and *Barbara* pictures:

1. Create a low-pass filter, (for instance) with the following parameters: type *ideal*, low frequency cut *0.3* (radius). Filter one of the images.
2. Create a high-pass filter that complements the low-pass (high frequency cut *0.7*). Filter the other image.

3. Merge both images in the frequency domain and perform the Inverse Fourier Transform.

4. Do the same but changing the filter operations (that is, if the low filter had been applied to *Mcqueen*, now it should be applied to *Barbara* and viceversa).

5. Visualize both results in a subplot from a close point of view with respect to the screen. Describe what can you see.

6. Walk back 4-5 meters and observe the display again. What happened? Why?